

Im base al numero di link entranti:

ORDINAMENTO

④	3 link
③	2 link
②	1 link
①	1 link

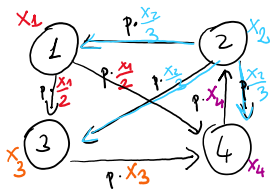
x_1 = importanza di ①
 x_2 = importanza di ②
 x_3 = importanza di ③
 x_4 = importanza di ④

DEFINIZIONE
 $\delta = \frac{1-p}{4}$
 importanza minima di ciascuna pagina

$p = 0.85$ $p \in (0,1)$
 ingegner

85% si trasferisce da una pagina all'altra

$p \cdot x_1$ percentuale di importanza che si trasferisce
 $\frac{p \cdot x_1}{2} \rightarrow ③$ $\frac{p \cdot x_1}{2} \rightarrow ④$



$$\begin{cases} x_1 = p \cdot \frac{x_2}{3} + \delta \\ x_2 = p \cdot x_4 + \delta \\ x_3 = p \cdot \frac{x_1}{2} + p \cdot \frac{x_2}{3} + \delta \\ x_4 = p \cdot \frac{x_1}{2} + p \cdot \frac{x_2}{3} + p \cdot x_3 + \delta \end{cases}$$

$p = 0.85$

$$\delta = \frac{1-p}{4}$$

MATRICE DEI COEFFICIENTI VETTORE TERMINE NOTO

$$\begin{cases} x_1 - 3x_2 = 7 \\ x_1 + 2x_2 = 9 \end{cases} \quad A = \begin{bmatrix} 1 & -3 \\ 1 & 2 \end{bmatrix} \quad b = \begin{bmatrix} 7 \\ 9 \end{bmatrix}$$

CRAMER

$$x_1 = \frac{\det \begin{pmatrix} 7 & -3 \\ 9 & 2 \end{pmatrix}}{\det \begin{pmatrix} 1 & -3 \\ 1 & 2 \end{pmatrix}} = \frac{14 + 27}{2 + 3} = \frac{41}{5} = 8.2$$

$$x_2 = \frac{\det \begin{pmatrix} 1 & 7 \\ 1 & 9 \end{pmatrix}}{\det \begin{pmatrix} 1 & -3 \\ 1 & 2 \end{pmatrix}} = \frac{9 - 7}{5} = \frac{2}{5} = 0.4$$

IN MATLAB

>> A=[1 -3; 1 2]

A =

1 -3
1 2

>> b=[7;9]

b =

7
9

>> x=A\b

x =

8.2000
0.4000

$p = 0.85$

$$\delta = \frac{1-p}{4}$$

$$\begin{cases} x_1 = p \cdot \frac{x_2}{3} + \delta \\ x_2 = p \cdot x_4 + \delta \end{cases}$$

$$x_1 = 0.1318 \dots$$

$$x_2 = 0.2222$$

$$\begin{cases} x_1 = \frac{1}{3} + \delta \\ x_2 = p \cdot x_4 + \delta \\ x_3 = p \cdot \frac{x_1}{2} + p \cdot \frac{x_2}{3} + \delta \\ x_4 = p \cdot \frac{x_1}{2} + p \cdot \frac{x_2}{3} + p \cdot x_3 + \delta \end{cases}$$

$$\begin{aligned} x_1 &= 0.1318... \\ x_2 &= 0.3329... \\ x_3 &= 0.1878... \\ x_4 &= 0.3475... \end{aligned}$$

```

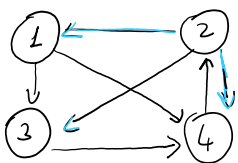
1 n=4; p=0.85; delta=(1-p)/n;
2 A=[1 -p/3 0 0
3     0 1 0 -p
4     -p/2 -p/3 1 0
5     -p/2 -p/3 -p 1]
6 b=[delta
7     delta
8     delta
9     delta]
10 x=A\b
  
```

Command Window output:

```

b =
    0.0375
    0.0375
    0.0375
    0.0375

x =
    0.1318
    0.3329
    0.1878
    0.3475
  
```



Im base al numero di link entranti:
ORDINAMENTO

4	3 link
3	2 link
2	1 link
1	

Page RANK
REDINAMENTO

4
2
3
1

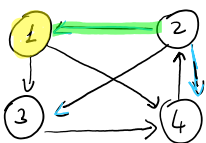
Come costruire in generale A b

MATRICE DI CONNETTIVITA'

o Numeriamo le pagine del web da 1 a n, e definiamo la matrice di connettività $G = (g_{ij})$, con

$g_{ij} = 1$ se c'è un link dalla pagina (i) alla pagina (j)
 $g_{ij} = 0$ altrimenti.

g_{ij}



numero di link che entrano in ciascuna pagina:

$$x_1 = g_{11} + g_{12} + g_{13} + g_{14} = 1$$

$$x_2 = g_{21} + g_{22} + g_{23} + g_{24} = 1$$

$$x_3 = g_{31} + g_{32} + g_{33} + g_{34} = 2$$

$$x_4 = g_{41} + g_{42} + g_{43} + g_{44} = 3$$

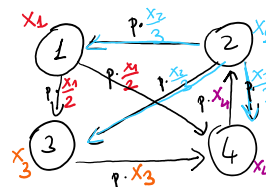
numero di link uscenti da ciascuna pagina:

$$c_1 = g_{11} + g_{21} + g_{31} + g_{41} = 2$$

$$c_2 = g_{12} + g_{22} + g_{32} + g_{42} = 3$$

$$c_3 = g_{13} + g_{23} + g_{33} + g_{43} = 1$$

$$c_4 = g_{14} + g_{24} + g_{34} + g_{44} = 1$$



$$\begin{cases} x_1 = p \cdot \frac{x_2}{3} + \delta \\ x_2 = p \cdot x_4 + \delta \end{cases}$$



$$\begin{cases} x_1 = p \cdot \left(g_{11} \frac{x_1}{c_1} + g_{12} \frac{x_2}{c_2} + g_{13} \frac{x_3}{c_3} + g_{14} \frac{x_4}{c_4} \right) + \delta \\ x_2 = p \cdot \left(g_{21} \frac{x_1}{c_1} + g_{22} \frac{x_2}{c_2} + g_{23} \frac{x_3}{c_3} + g_{24} \frac{x_4}{c_4} \right) + \delta \end{cases}$$

$$\begin{cases} x_1 = p \cdot \frac{x_2}{3} + \delta \\ x_2 = p \cdot x_4 + \delta \\ x_3 = p \cdot \frac{x_1}{2} + p \cdot \frac{x_2}{3} + \delta \\ x_4 = p \cdot \frac{x_1}{2} + p \cdot \frac{x_2}{3} + p \cdot x_3 + \delta \end{cases}$$

$$\begin{cases} x_2 = p \cdot \left(g_{21} \frac{x_1}{c_1} + g_{22} \frac{x_2}{c_2} + g_{23} \frac{x_3}{c_3} + g_{24} \frac{x_4}{c_4} \right) + \delta \\ x_3 = p \cdot \left(g_{31} \frac{x_1}{c_1} + g_{32} \frac{x_2}{c_2} + g_{33} \frac{x_3}{c_3} + g_{34} \frac{x_4}{c_4} \right) + \delta \\ x_4 = p \cdot \left(g_{41} \frac{x_1}{c_1} + g_{42} \frac{x_2}{c_2} + g_{43} \frac{x_3}{c_3} + g_{44} \frac{x_4}{c_4} \right) + \delta \end{cases}$$

IN GENERALE

$$x_i = p \cdot \left(\sum_{j=1}^m g_{ij} \frac{x_j}{c_j} \right) + \delta \quad i=1, \dots, m$$

$$A = \begin{pmatrix} 1 - p \frac{g_{11}}{c_1} & -p \frac{g_{12}}{c_2} & -p \frac{g_{13}}{c_3} & -p \frac{g_{14}}{c_4} \\ -p \frac{g_{21}}{c_1} & 1 - p \frac{g_{22}}{c_2} & -p \frac{g_{23}}{c_3} & -p \frac{g_{24}}{c_4} \\ -p \frac{g_{31}}{c_1} & -p \frac{g_{32}}{c_2} & 1 - p \frac{g_{33}}{c_3} & -p \frac{g_{34}}{c_4} \\ -p \frac{g_{41}}{c_1} & -p \frac{g_{42}}{c_2} & -p \frac{g_{43}}{c_3} & 1 - p \frac{g_{44}}{c_4} \end{pmatrix} \quad b = \begin{pmatrix} \delta \\ \delta \\ \delta \\ \delta \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} - p \cdot \begin{pmatrix} \frac{g_{11}}{c_1} & \frac{g_{12}}{c_2} & \frac{g_{13}}{c_3} & \frac{g_{14}}{c_4} \\ \frac{g_{21}}{c_1} & \frac{g_{22}}{c_2} & \frac{g_{23}}{c_3} & \frac{g_{24}}{c_4} \\ \frac{g_{31}}{c_1} & \frac{g_{32}}{c_2} & \frac{g_{33}}{c_3} & \frac{g_{34}}{c_4} \\ \frac{g_{41}}{c_1} & \frac{g_{42}}{c_2} & \frac{g_{43}}{c_3} & \frac{g_{44}}{c_4} \end{pmatrix}$$

$$A = I - p \cdot G \cdot D$$

$G \cdot D$

$$D = \begin{pmatrix} \frac{1}{c_1} & 0 & 0 & 0 \\ 0 & \frac{1}{c_2} & 0 & 0 \\ 0 & 0 & \frac{1}{c_3} & 0 \\ 0 & 0 & 0 & \frac{1}{c_4} \end{pmatrix}$$

$$G = \begin{pmatrix} g_{11} & g_{12} & g_{13} & g_{14} \\ g_{21} & g_{22} & g_{23} & g_{24} \\ g_{31} & g_{32} & g_{33} & g_{34} \\ g_{41} & g_{42} & g_{43} & g_{44} \end{pmatrix}$$

IN GENERALE

$(GD)_{ij}$

$$x_i = p \cdot \left(\sum_{j=1}^m \frac{g_{ij}}{c_j} x_j \right) + \delta \quad i=1, \dots, m$$

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} \quad e = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$$

formulazione
vettoriale

$$x = p(GD) \cdot x + \delta e$$

$$\delta \cdot e = \delta \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$$

• versione $\rightarrow x = 1 \quad \dots$

$$\underbrace{(\mathbb{I} - p(G))}_A x = \underbrace{\delta e}_b$$

$$\delta \cdot e = \delta \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} \vdots \\ \delta \end{pmatrix}$$